

PRESERVING RARE-CLASS SIGNAL IN FEDERATED LEARNING

Under Statistical and System Heterogeneity

Barbara Koch | MPhil Data Intensive Science | University of Cambridge

Why This Project Matters

Accurate diagnostic models need large, diverse image collections, but in medicine those images are scattered across hospitals and usually cannot be pooled because of privacy law. *Federated learning* (FL) offers a way out [1]: each site keeps its data and shares only model updates, which a central server merges into one global model. Two practical problems make this hard in medicine. Sites see different case mixes (*statistical heterogeneity*), and they differ in compute and availability (*system heterogeneity*), so some sites return only partial work in a given round. Medical image data is also severely class-imbalanced: a few benign conditions dominate, while the urgent diagnoses are rare. The worst case is when the slow or under-resourced site is the one holding the rare disease. A model can then report high overall accuracy while quietly failing on exactly the lesions that matter clinically. Crucially, the information needed to recognise a rare disease reaches the shared model only through the updates of the site that holds it — so the algorithm that *merges* those updates, not the network architecture, decides whether that rare-disease signal survives (Figure 1).

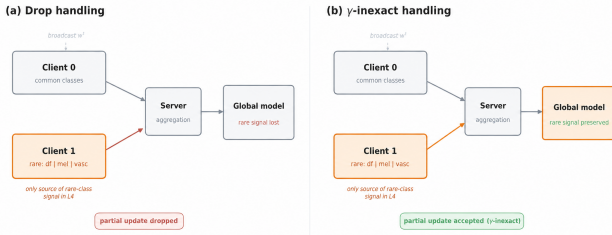


Figure 1: How a slow rare-class site’s update is handled. Discarding the partial update (a) loses the rare-disease signal; accepting it (b) lets that signal reach the global model — the mechanism behind the headline result.

Approach

We studied this on DermaMNIST [2], a seven-class dermatology benchmark of small (28×28) skin-lesion images [3] in which benign moles make up two-thirds of the data and the urgent classes, including melanoma, are rare. Using controlled, simulated federations, we compared the three most widely used FL merging algorithms — FedAvg [4], FedProx [5] and FedNova [6] — and asked not “which is best?” but “in which conditions does each preserve or lose rare-disease performance?” We scored models by *macro-F1*, which weights all seven classes equally and therefore exposes the rare-class failure that raw accuracy hides. We then varied one factor at a time: how data is split across sites, whether a slow site’s partial update is discarded or accepted, how strongly a site’s learning is weakened, and whether the training loss is rebalanced. The sharpest test used two matched two-client federations differing in only one respect — whether the slow site

uniquely holds the rare diseases — so any performance gap cleanly isolates the value of admitting that unique signal. A single idea organises the work — *rare-client signal flow*: a rare disease is learned only to the extent that the holding site’s contribution reaches and persists in the global model.

Key Findings

Differences between the algorithms are small until the right stress is applied (Table 1). With balanced participation, FedProx and FedAvg differ by under 0.01 macro-F1, and FedProx’s one tuning knob helps only in a narrow range — pushed an order of magnitude too high it costs 0.074 macro-F1 and damages precisely the rare classes. The central result concerns slow sites (“stragglers”). When the rare-disease site is the straggler, the FedProx + γ -inexact protocol beats FedAvg + drop by a large +0.115 macro-F1; but a controlled decomposition shows that about 96% of that gain (+0.111) traces to the γ -inexact handling that *accepts* the slow site’s partial update — the protocol conventionally paired with FedProx here — and almost none (+0.004) to the proximal regularisation term itself. A matched control in which the slow site holds nothing unique erases the effect, confirming the mechanism is the admission of rare-disease signal, not the algorithm’s branding. Under an extreme “perfect storm,” the FedProx + γ -inexact protocol reaches 0.49 macro-F1 where FedAvg + drop collapses to 0.09 (Figure 2). FedNova proved regime-dependent: robust when a site is steadily weakened (leading by +0.131 in the most extreme case) but collapsing to 0.24 under random, round-to-round slowness. Every failure shared one signature — rare lesions misrouted into the benign majority class (in one cell, melanoma was recovered 1% of the time by FedAvg versus 46% by FedNova). Finally, simply reweighting the training loss matched FedProx, showing that imbalance correction is a separate lever from update merging.

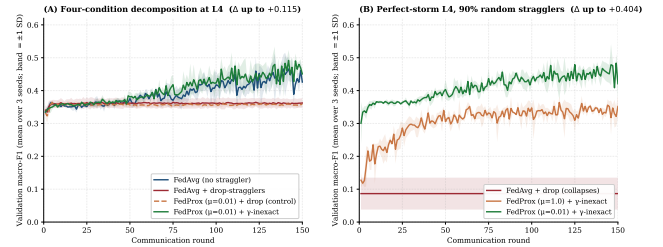


Figure 2: Validation macro-F1 over training at the two largest gaps. Under the FedProx + γ -inexact protocol the slow site’s partial update is accepted and rare-class performance recovers; the FedAvg + drop protocol discards it and flatlines.

Interpretation & Impact

All of these results reduce to one rule: rare-class performance is preserved precisely when rare-site signal reaches the global model with usable weight. This re-

Table 1: Regime map: which mechanism preserves rare-class signal, and the supporting macro-F1 evidence. There is no universal winner — the protective mechanism changes with the heterogeneity regime.

Heterogeneity regime	What preserves rare-class signal	Macro-F1 evidence
Balanced / IID participation	No method needed (all comparable)	$\Delta \approx 0$ (IID -0.007)
Statistical heterogeneity only	Weak FedProx edge	$\Delta \approx +0.007$
Straggler drops the rare-class site	Accepting the partial update (FedProx + γ -inexact)	$+0.115$ total; 96% from acceptance, 4% from the proximal term
“Perfect storm” (90% random stragglers)	Accepting partial updates	0.49 vs 0.09 when dropped
Steady learning-rate weakening	FedNova (work-normalised merging)	leads by $+0.131$ at 50:1
Random, round-to-round slowness	None robust; FedNova collapses	0.24 (majority-only)

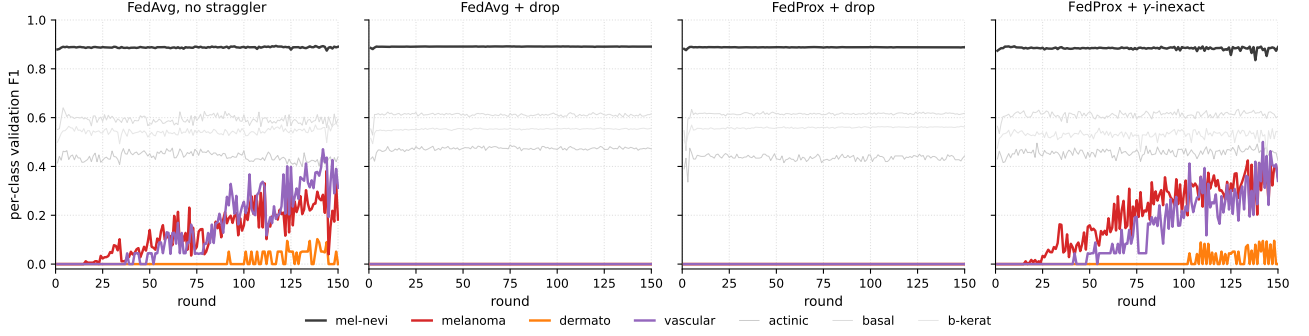


Figure 3: Per-class validation F1 under the four-condition decomposition. The majority class is unaffected, but the three rare classes flatline under *both* drop conditions and recover *only* when the partial update is accepted (FedProx + γ -inexact) — pinpointing rare-class learning as the locus of the effect.

frames a crowded methods debate. There is no universal best FL optimiser; the right choice is dictated by the heterogeneity regime, and the decisive factor is often a protocol detail — whether partial updates are admitted — rather than the headline algorithm. For anyone building imbalanced medical FL systems the practical guidance follows directly from the regime, summarised in the box below. The contribution is a regime map and a mechanism — conceptually relevant beyond dermatology, and worth testing in other rare-class federated settings.

Next Steps

The most promising extension is to combine partial-update acceptance with loss rebalancing, which were effective separately here. Beyond that: validate the map on higher-resolution, genuinely multi-site data with real wall-clock deadlines; broaden the comparison to variance-reduced and adaptive optimisers; and log per-site performance to quantify fairness across hospitals — but the mechanism isolated here is exactly the one a real deployment must protect.

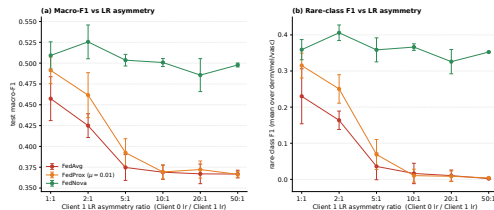


Figure 4: FedNova: robust to steady weakening, brittle under random stragglers.

What This Means in Practice

- *Near-IID clients* — FedAvg is enough.
- *Statistical heterogeneity alone* — the FedProx edge is small and fragile.
- *Rare-class site straggling* — accepting its partial update (FedProx + γ -inexact) is the key protection.
- *Rare site steadily weakened* — FedNova’s work-normalised merging can help.
- *Random round-to-round straggling* — do not assume FedNova is safe.
- *Reporting* — use macro-F1 and per-class/rare-class metrics, never accuracy alone.

Scope and caveats. A single 28×28 DermaMNIST benchmark; simulated heterogeneity; mechanism probes at $n = 3$ seeds; clinically motivated, not validated. The straggler gain is attributed to the FedProx + γ -inexact *protocol*, not to partial-update acceptance as a proven standalone mechanism.

References

- [1] N. Rieke et al. The future of digital health with federated learning. *npj Digital Medicine*, 3:119, 2020.
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